Experiment 04: Software Provisioning and Configuration using Ansible

Experiment No.	04
Mapped CO	CO4
Aim	To automate the provisioning and configuration of software on target hosts using Ansible.
Tools / Software	Ansible; One or more Linux hosts / VMs reachable over SSH; SSH key pair; Terminal / editor

Theory

Ansible is an agentless configuration-management and automation tool used to provision, configure and manage servers. It connects to managed nodes mainly through SSH and applies the desired system state written in YAML files. Since no agent is required on the target host, Ansible is simple to start with and suitable for repeatable software configuration tasks.

The main components of Ansible are inventory, modules, playbooks and roles. An inventory lists the target hosts, modules perform specific tasks such as installing a package or starting a service, and playbooks describe ordered tasks to be executed on selected hosts. Idempotence is an important property of Ansible: running the same playbook multiple times should not repeatedly change a system that is already in the desired state.

Procedure

1. Install Ansible on the control node and verify the installation using ansible --version.
2. Create or identify one Linux managed host / VM that is reachable from the control node using SSH.
3. Configure passwordless SSH access using an SSH key pair or ensure that valid SSH credentials are available.
4. Create an inventory file named inventory.ini and list the target host under a group such as [web].
5. Verify connectivity to the managed host using the Ansible ping module.
6. Write a playbook named site.yml to install and start a web server such as nginx and deploy a simple web page.
7. Run the playbook using ansible-playbook and observe the ok / changed / failed task summary.
8. Re-run the playbook to verify idempotence and confirm that no unnecessary changes are made.

Sample Code / Commands

inventory.ini

```
[web]
192.168.56.10 ansible_user=ubuntu
```

site.yml

```
- name: Configure web servers
  hosts: web
  become: true
  tasks:
    - name: Install nginx
      apt:
        name: nginx
        state: present
```

```
        update_cache: true

    - name: Ensure nginx is running
      service:
        name: nginx
        state: started
        enabled: true

    - name: Deploy index page
      copy:
        content: "<h1>Welcome to Ansible Lab</h1>"
        dest: /var/www/html/index.html
```

Run commands

```
ansible --version
ansible -i inventory.ini web -m ping
ansible-playbook -i inventory.ini site.yml
ansible-playbook -i inventory.ini site.yml
```

Observation / Experiment Output

Instructions to students: Paste screenshots or copied outputs in the boxes below after completing Ansible inventory creation, connectivity testing, playbook execution and verification. Keep screenshots clear and readable.

Name	Roll No.	Batch	Date
Gargee Warke	65	B2	14/08/26

1. Ansible Installation and Version Output

```
en_PH.utf8
en_SG.utf8
en_US.utf8
en_ZA.utf8
en_ZM.utf8
en_ZW.utf8
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ export LANG=en_IN.UTF-8
export LANGUAGE=en_IN:en
export LC_ALL=en_IN.UTF-8
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ locale
LANG=en_IN.UTF-8
LANGUAGE=en_IN:en
LC_CTYPE="en_IN.UTF-8"
LC_NUMERIC="en_IN.UTF-8"
LC_TIME="en_IN.UTF-8"
LC_COLLATE="en_IN.UTF-8"
LC_MONETARY="en_IN.UTF-8"
LC_MESSAGES="en_IN.UTF-8"
LC_PAPER="en_IN.UTF-8"
LC_NAME="en_IN.UTF-8"
LC_ADDRESS="en_IN.UTF-8"
LC_TELEPHONE="en_IN.UTF-8"
LC_MEASUREMENT="en_IN.UTF-8"
LC_IDENTIFICATION="en_IN.UTF-8"
LC_ALL=en_IN.UTF-8
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ ansible --version
ansible [core 2.14.18]
  config file = None
  configured module search path = ['/home/apsit/.ansible/plugins/modules', '/usr/share/
ansible/plugins/modules']
  ansible python module location = /home/apsit/.local/pipx/venvs/ansible-core/lib/pytho
n3.10/site-packages/ansible
  ansible collection location = /home/apsit/.ansible/collections:/usr/share/ansible/col
lections
  executable location = /home/apsit/.local/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0] (/home/apsit/.loc
al/pipx/venvs/ansible-core/bin/python)
  jinja version = 3.1.6
  libyaml = True
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$
```

```
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC: ~/ansible-nginx-de...
apsit@apsit-HP-Pro-Towe...  apsit@apsit-HP-Pro-Towe...  apsit@apsit-HP-Pro-Tower...
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-136-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.


Expanded Security Maintenance for Applications is not enabled.

6 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

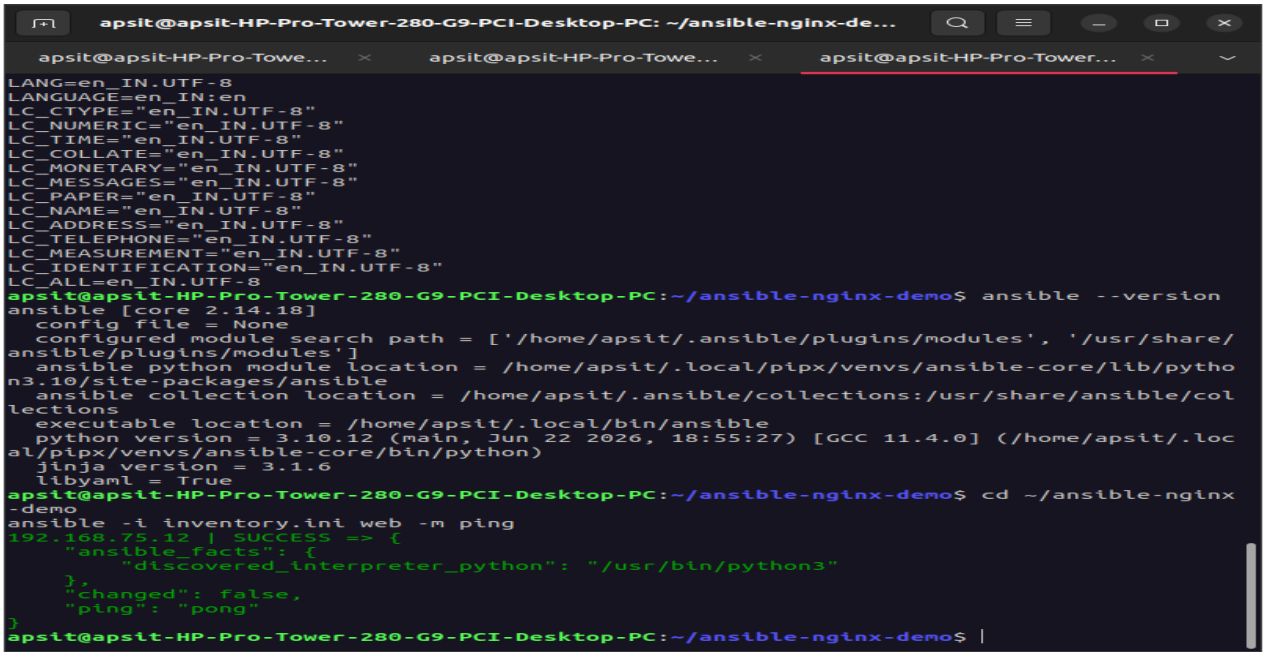
32 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

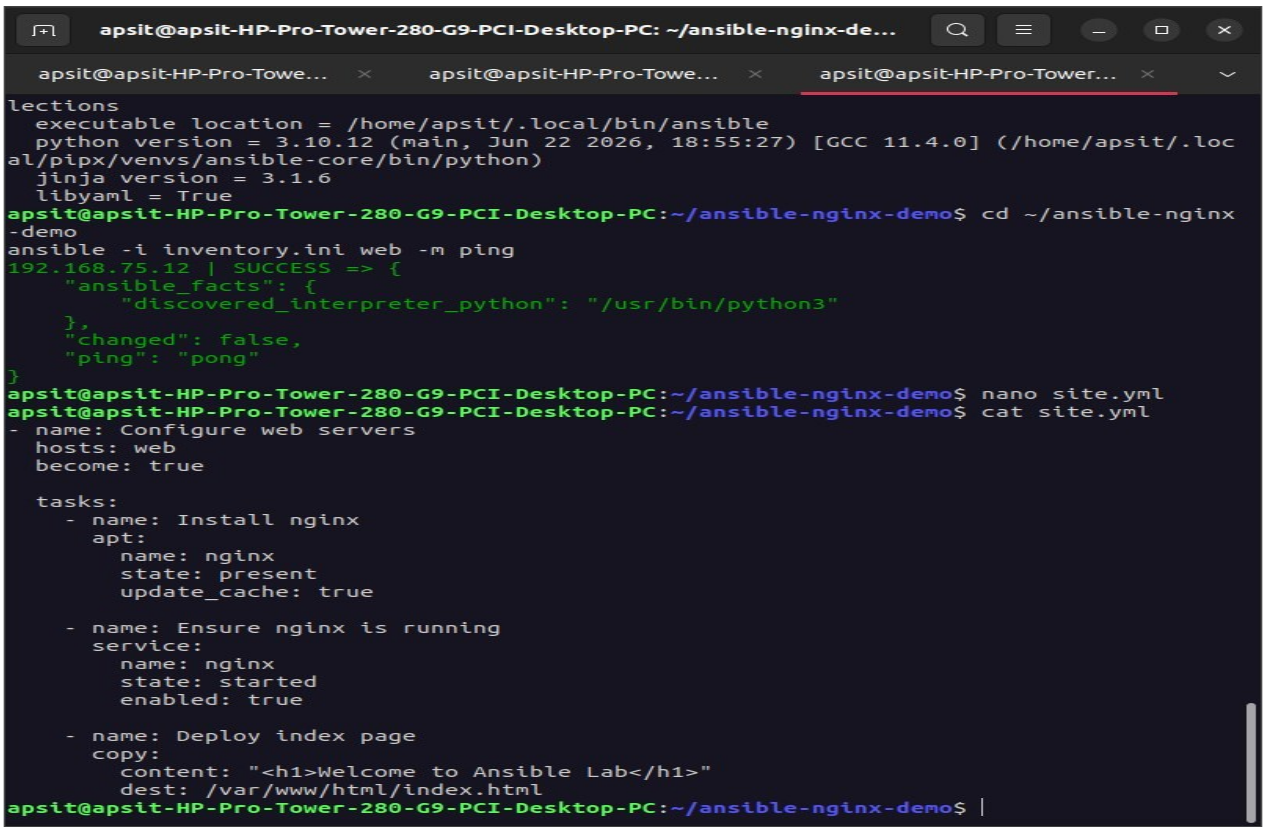
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


1 device has a firmware upgrade available.
Run `fwupdmgr get-upgrades` for more information.

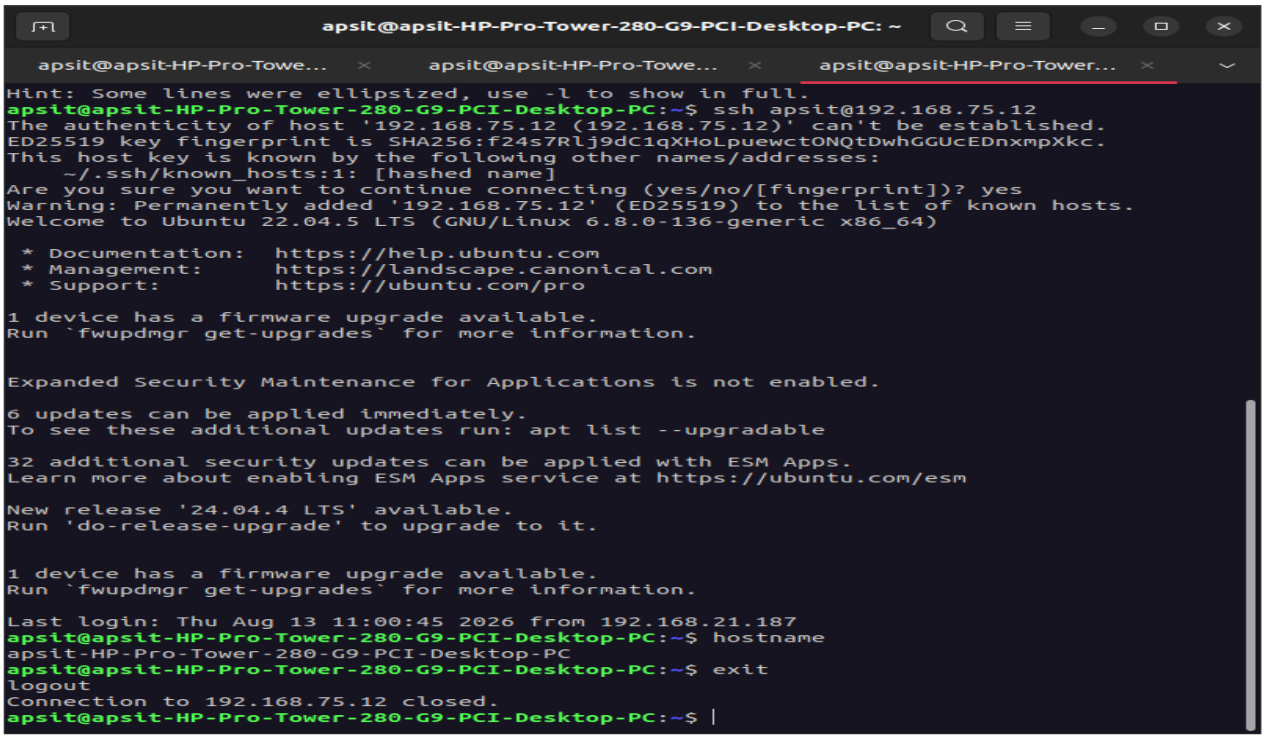
Last login: Fri Aug 14 08:23:49 2026 from 192.168.75.12
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ hostname
apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ exit
logout
Connection to 192.168.75.12 closed.
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~$ mkdir -p ~/ansible-nginx-demo
cd ~/ansible-nginx-demo
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ pwd
/home/apsit/ansible-nginx-demo
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ nano inventory.ini
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$ cat inventory.ini
[web]
192.168.75.12 ansible_user=apsit
apsit@apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC:~/ansible-nginx-demo$
```

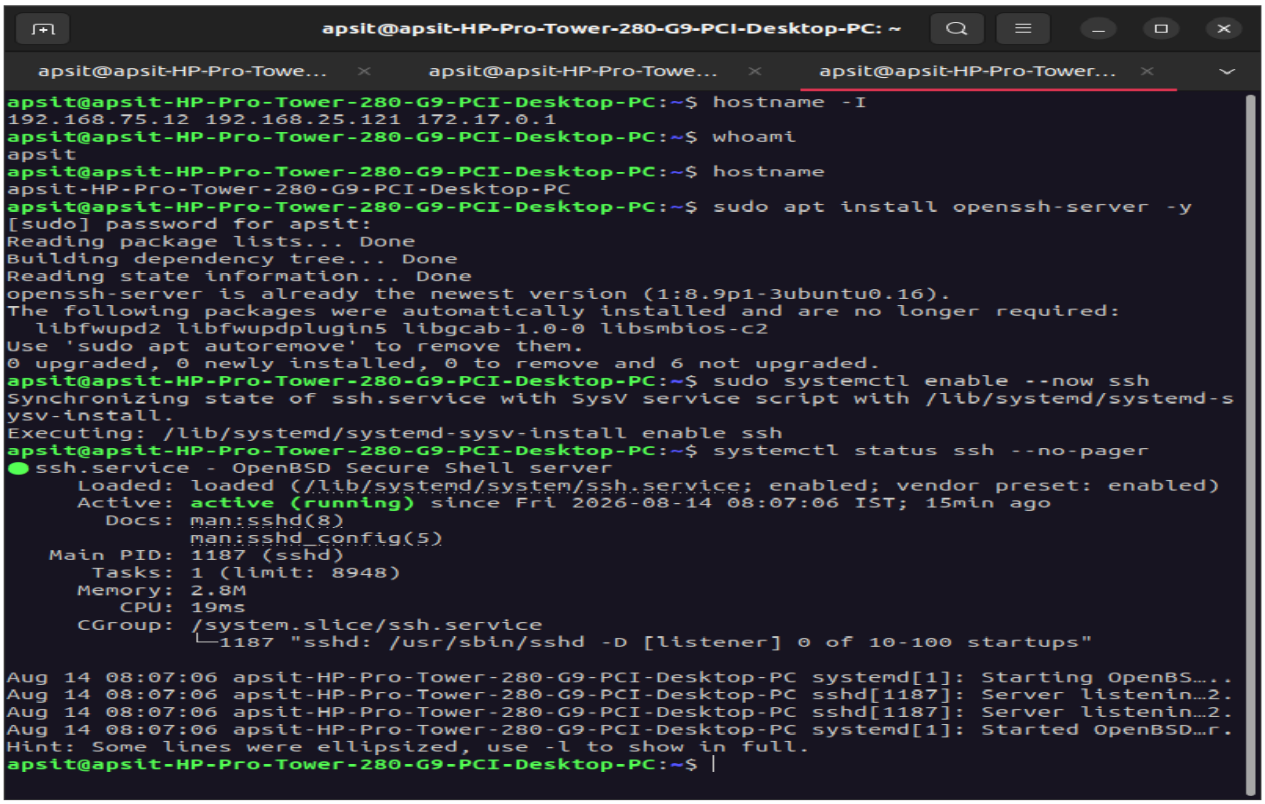
2. Inventory File Output



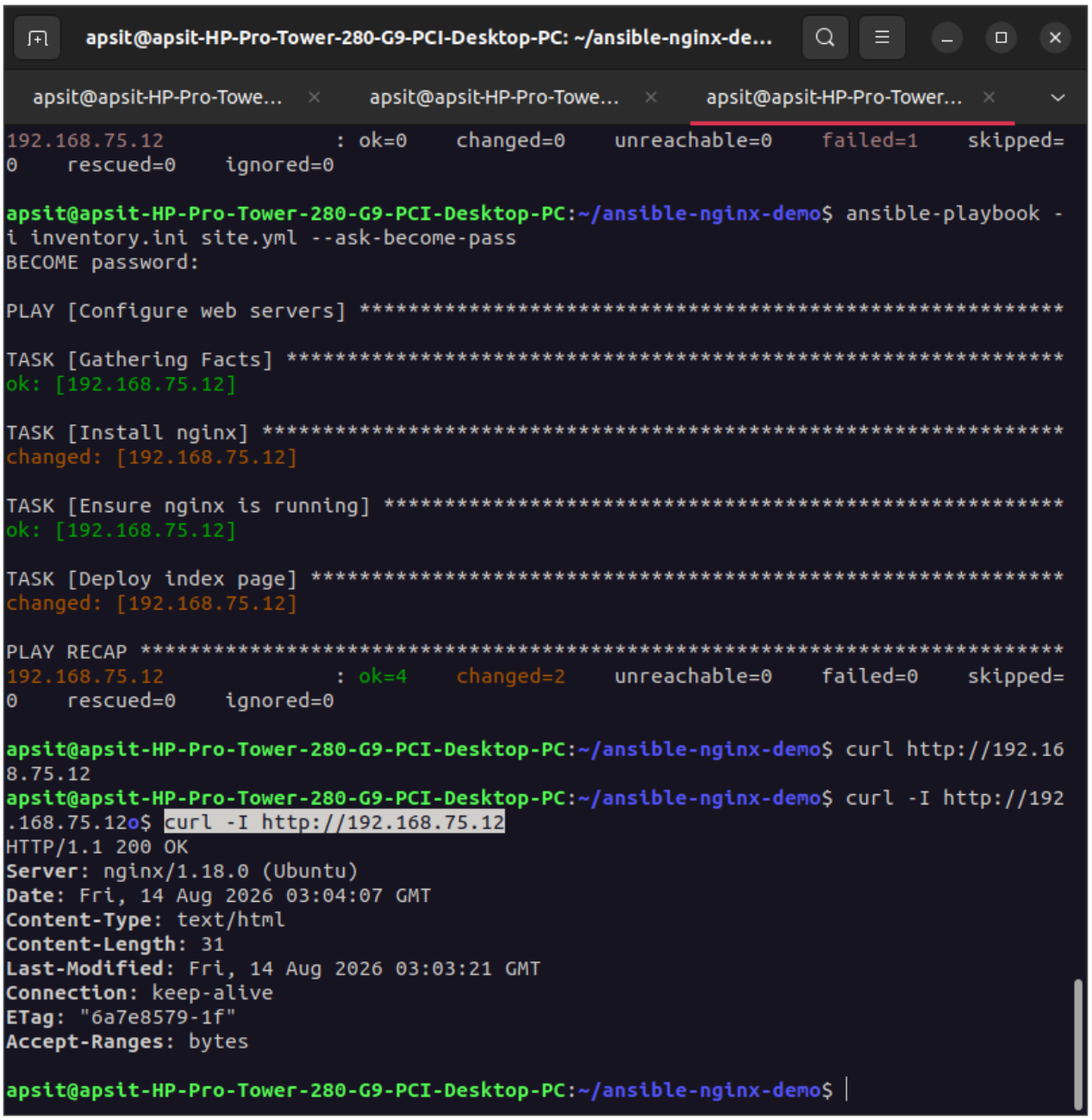


3. SSH Connectivity / Ping Module Output

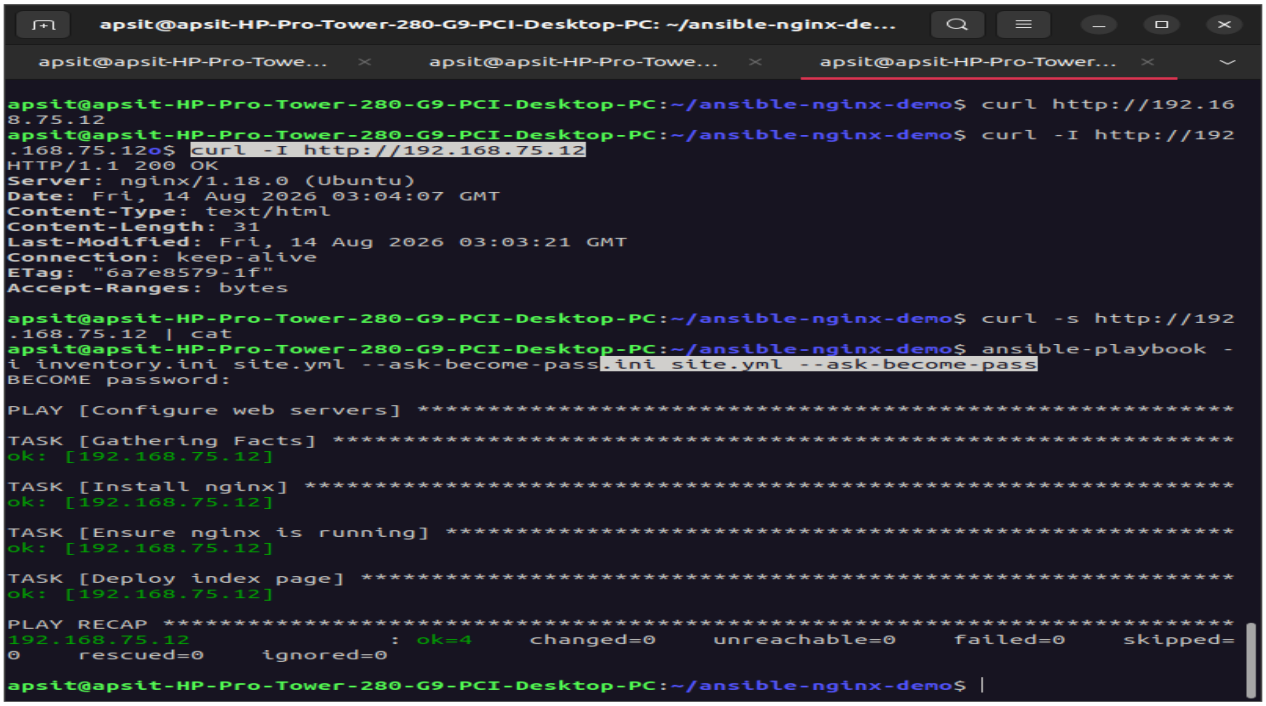


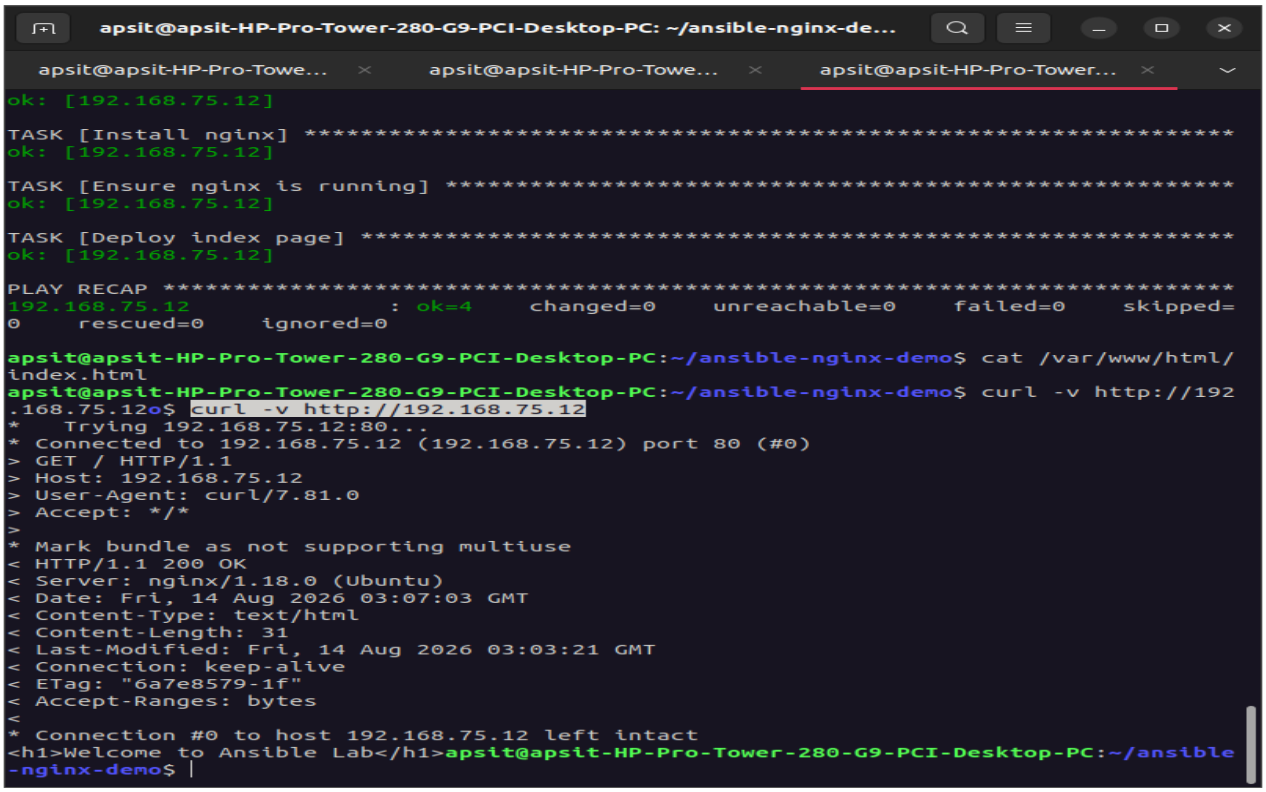


4. Playbook File Output



5. Playbook Execution and Idempotence Output





GitHub Repository / Lab Folder Link:
Local Lab Folder: /home/apsit/ansible-nginx-demo

Managed Host / VM Details:
Managed Host IP: 192.168.75.12
Username: apsit
Operating System: Ubuntu 22.04.5 LTS
Hostname: apsit-HP-Pro-Tower-280-G9-PCI-Desktop-PC
SSH: Passwordless SSH configured
Web Server: nginx

Final Observation Statement:

Ansible was successfully used to connect to the managed Ubuntu host through SSH and automate nginx installation, service configuration, and web-page deployment. The nginx page was successfully verified using curl with HTTP 200 OK. On the second playbook execution, changed=0 was observed, demonstrating idempotent configuration management.

Expected Output

The managed host should have nginx installed and running. The web server should serve the deployed index page, the Ansible playbook should complete successfully, and the second run should show idempotent behaviour with no unnecessary changes.

Conclusion

Software provisioning and configuration were automated declaratively with Ansible, demonstrating agentless management, repeatable configuration and idempotent convergence to the desired system state.

Viva / Review Questions

1. Why is Ansible described as agentless?
2. What does idempotence mean in configuration management?
3. Differentiate a playbook from a role.
4. What transport does Ansible use to reach Linux managed nodes?

Marks: ____/ 10 Date: _____ Signature of Faculty: _____